

Variational Autoencoders for Missing Data Imputation with Application to a Simulated Milling Circuit

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Abstract: Missing data values and differing sampling rates, particularly for important parameters such as particle size and stream composition, are a common problem in minerals processing plants. Missing data imputation is used to avoid information loss (due to downsampling or discarding incomplete records). A recent deep-learning technique, variational autoencoders (VAEs), has been used for missing data imputation in image data, and was compared here to imputation by mean replacement and by principal component analysis (PCA) imputation. The techniques were compared using a synthetic, nonlinear dataset, and a simulated milling circuit dataset, which included process disturbances, measurement noise, and feedback control. Each dataset was corrupted with missing values in 20% of records (lightly corrupted) and in 90% of records (heavily corrupted). For both lightly and heavily corrupted datasets, the root mean squared error of prediction for VAE imputation was lower than the traditional methods. Possibilities for the extension of missing data imputation to inferential sensing are discussed.

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1 INTRODUCTION

Large quantities of historical data are available for most industrial operations, which may contain valuable information to be used for developing monitoring methods, training soft sensors, and tuning controllers (He and Wang, 2017; Qin, 2014). However, industrial data records frequently contain missing values (due to equipment failures, especially in harsh minerals processing plants), and have differing sampling rates for variables, due to online and offline measurements, sampling delays or equipment limitations (Bergh, 2010).

The result is a reduction in the amount of information available, as data is frequently downsampled to match the measurement with the lowest frequency, or records with missing values are discarded (Kadlec et al., 2009). Missing value imputation is a strategy to replace missing values with plausible values, in order to reduce data loss (Kiers, 1997; Sedghi et al., 2017; Zeng et al., 2010). Imputation of missing values could result in more uptime for controllers, soft sensors and monitoring methods, and upsampling of low frequency measurements could result in faster control, improving plant performance.

Variational autoencoders (VAEs) are a recently proposed deep learning-based generative model, which can be used for unsupervised learning of the distribution of a dataset, and the generation of further samples which closely resemble the original data (Kingma and Welling, 2013; Rezende et al., 2014). These samples could be used for testing soft sensors, controllers and monitoring methods. VAEs have also been used for missing data imputation in image-based data, suggesting the potential for application to process data.

This paper compares missing data imputation by traditional methods with imputation by VAE. Section 2 gives the background for imputation and VAEs, and Section 3 describes the datasets and imputation methods. The results are shown in Section 4, followed by conclusions in Section 5.

2 BACKGROUND

2.1 Missing data imputation approaches

The three most common approaches to deal with missing values are deletion of records with missing values, single imputation (such as replacement by the mean) and multiple imputation, which makes use of iterative model-based methods (Kadlec et al., 2009). A common multiple imputation method is maximum likelihood iterative principal component analysis (PCA) imputation (Dray and Josse, 2015; Kiers, 1997). In the process/chemical engineering field, the method has been applied to imputation for a simulated flow network (Imtiaz et al., 2004).

A number of other machine learning methods have been used for model-based multiple imputation in process engineering applications. Supervised decision trees have been successfully applied as the regression model for multiple imputation of industrial maintenance databases (Lakshminarayan et al., 1999) and of missing values in ironmaking blast furnaces (Zeng and Gao, 2009). Mean replacement, PCA imputation, k-nearest neighbours regression imputation and neural network imputation were compared in the context of fault diagnosis of the Tennessee Eastman benchmark problem and an industrial gas sweetening plant (Askarian et al., 2016).

2.2 Variational autoencoders

A variational autoencoder (VAE) makes use of a similar architecture to a standard autoencoder, which is shown in Fig. 1a). Autoencoders learn a latent space representation of input data, typically for dimensionality reduction and feature extraction (Hinton and Salakhutdinov, 2006; Yan et al., 2017), or to reconstruct a denoised version of the input (Gondara and Wang, 2017; Vincent et al., 2008).

However, whereas an autoencoder learns a latent space representation of the data, a VAE learns a **distribution** in that latent space representation. The result is that samples can be drawn from the latent space distribution, input to the decoder network, and new samples of data generated.

The underlying assumption of the VAE is that the data X is generated by some underlying distribution $p(X)$, and that this unknown distribution can be represented by the latent variable, Z , which is itself generated by some distribution $p(Z)$. The joint distribution, $p(X, Z)$ can then be represented as:

$$p(X, Z) = p_\theta(X|Z)p(Z) \quad (1)$$

In equation (1), the joint distribution can be generated by sampling from the distribution of Z , $p(Z)$, also known as the prior, and the distribution of X given Z , also known as the likelihood $p_\theta(X|Z)$, which is parameterised by θ . The likelihood function, with parameters θ , can be represented by a neural network, which is analogous to the decoder network in a standard autoencoder. The parameters, θ , must be learnt from the data, as will be described below.

The prior, $p(Z)$, is typically chosen to be the multi-dimensional zero-centred normal distribution with unit variance (Kingma and Welling, 2013), and thus has no parameters to learn.

The marginal likelihood of the model, $p_\theta(X)$, of the unknown distribution, $p(X)$, can be obtained by integration of equation (1), specifically the likelihood and prior. The marginal likelihood (or evidence) specifies how likely the data, X , is under the model with parameters θ .

$$p_\theta(X) = \int p(X, Z) dZ = \int p_\theta(X|Z)p(Z) dZ \quad (2)$$

From Bayes' rule, the marginal likelihood $p_\theta(X)$ can be related to the posterior distribution $p_\theta(Z|X)$.

$$p_\theta(Z|X) = \frac{p_\theta(X|Z)p(Z)}{p_\theta(X)} \quad (3)$$

The posterior is the distribution of the latent variable Z , given observations of the data X , but it cannot be determined analytically. The posterior is therefore approximated by $q_\phi(Z|X)$, typically a neural network with parameters ϕ , which is analogous to the encoder network in a standard autoencoder.

The best representation (in other words, the lowest reconstruction error) of the data X will be obtained by maximising the logarithm of the marginal likelihood (the log-evidence), $\ln p_\theta(X)$, by changing the parameters of the encoder network, ϕ , and of the decoder network, θ . It can be shown (refer to Kingma and Welling (2013) for details) that

the log-evidence can be expressed in terms of the “evidence lower bound”, or ELBO:

$$\ln p_\theta(X) = ELBO + KL[q_\phi(Z|X)||p_\theta(Z|X)] \quad (4)$$

In equation (4), the second term on the right hand side is the Kullback-Liebler divergence, which describes the agreement between two distributions, in this case the approximation to the posterior distribution, or the encoder network, $q_\phi(Z|X)$, and the true posterior distribution, $p_\theta(Z|X)$. The KL divergence is zero if the two distributions are identical (in other words, if the approximation of the posterior is perfect), and positive for non-identical distributions. Since the KL divergence is always non-negative, the log-evidence is always greater than or equal to the ELBO:

$$\ln p_\theta(X) \geq ELBO \quad (5)$$

Thus, the log-evidence can be driven towards its maximum value by finding the parameters ϕ and θ which maximise the ELBO, defined in equation (6).

$$ELBO = E_{q_\phi(Z|X)}[\ln p_\theta(X|Z)] - KL[q_\phi(Z|X)||p(Z)] \quad (6)$$

In equation (6), the first term is the expectation of the log-likelihood of the decoder network, given the encoder network's output. Maximising this term corresponds to minimising the reconstruction error of a standard autoencoder.

The second term in equation (6) is the Kullback-Liebler divergence between the approximated posterior distribution, or the encoder network, $q_\phi(Z|X)$, and the prior distribution of the latent variable Z , $p(Z)$. Minimising this term brings the approximate posterior, $q_\phi(Z|X)$, closer to the prior, $p(Z)$, which has the effect of regularisation, preventing overfitting.

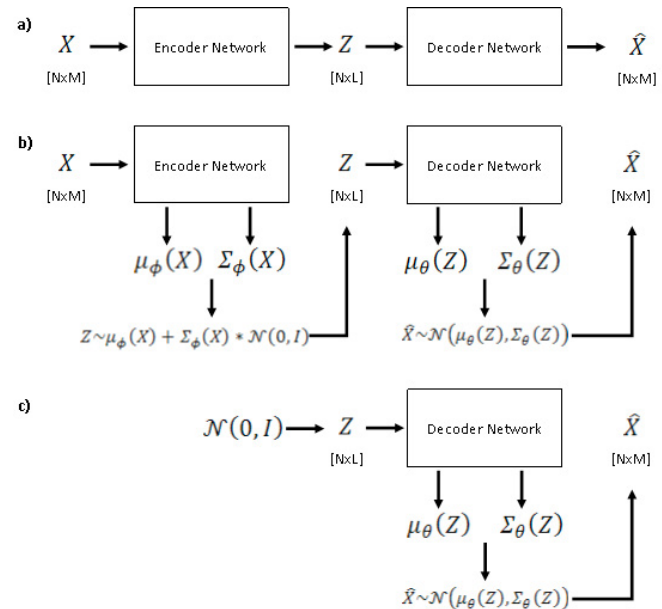


Fig. 1. Structure of autoencoders: a) Standard autoencoder; b) Variational autoencoder during training and imputation; c) Generative variational autoencoder.

2.3 VAE Structure

The structure of the VAE used here is shown in Fig. 1b). Given data X , the encoder network outputs the mean and variance of a multidimensional Gaussian which describes the latent space distribution, or approximate posterior distribution, $q_\phi(Z|X)$. Sampling from this distribution generates values for Z .

Given Z , the decoder network outputs the mean and variance of the multidimensional Gaussian describing the reconstructed data space, the likelihood $p_\theta(X|Z)$. Sampling from this distribution generates $\hat{X}$.

Due to the definition of the evidence lower bound in equation (6), training the VAE forces the encoder network's output distribution close to the prior distribution of Z , $p(Z)$, which is defined as $\mathcal{N}(0, I)$. Thus, it is possible to discard the encoder network of the VAE, and generate new data by drawing a sample from $\mathcal{N}(0, I)$, and feeding this into the decoder network, as shown in Fig. 1c).

3 METHODOLOGY

In order to demonstrate an application of variational autoencoders (VAEs), missing data imputation by VAE will be compared to single imputation and multiple imputation by Principal Component Analysis, using a synthetic dataset and a simulated milling circuit dataset.

3.1 Single imputation

Single imputation is achieved by the replacement of missing values with the mean of the observed values for each variable.

3.2 Multiple imputation by PCA

Multiple imputation using PCA is achieved using the approach as summarised below (Dray and Josse, 2015; Kiers, 1997):

1. Replace missing values with random values;
2. Build a model of the data:
 - a. Standardise the data;
 - b. Find the principal components and eigenvalues by singular value decomposition;
 - c. Find retained variance from eigenvalues;
 - d. Determine number of principal components which retains required variance;
3. Reconstruct missing and observed values with the model:
 - a. Reconstruct the data using the retained principal components;
 - b. Convert the standardised data back to the original data space using the standardisation parameters in 2.a. above.
4. Replace the missing values with the reconstructed values, leaving the observed values unchanged;
5. Compute the reconstruction error of the observed values;
6. If the reconstruction error is below a specified tolerance, or the iteration limit has been reached, end.
7. Otherwise, return to step 2.

Three hyperparameters must be defined for PCA imputation: the maximum number of iterations, the tolerance on reconstruction error, and the number of principal components to retain (specified here in terms of explained variance).

3.3 Multiple imputation by VAE

Missing data imputation is performed with a trained VAE using a Markov chain Monte Carlo method, which has been shown to converge to the true marginal distribution of missing values, given the observed values (Rezende et al., 2014). In practice, the imputation procedure is similar to that for PCA:

1. Replace missing values in data with random values;
2. Input the data to the trained VAE:
 - a. Sample from the latent variable distribution (the output of the encoder) to generate Z , given X ;
 - b. Sample from the reconstructed data distribution (the output of the decoder) to generate $\hat{X}$, given Z .
3. Replace the missing values with the reconstructed values, leaving the observed values unchanged;
4. Compute the reconstruction error of the observed values;
5. If the reconstruction error is below a specified tolerance, or the iteration limit has been reached, end.
6. Otherwise, return to step 2.

The hyperparameters which must be specified include: network architecture and neuron activations (for encoding and decoding networks), training algorithm (including optimiser and learning rate), the dimensionality of the latent space, the maximum training and imputation iterations, batch size, and tolerance on reconstruction error.

3.4 Datasets

Missing data imputation is performed on a synthetic, nonlinear dataset, and on a simulated milling circuit dataset.

Synthetic dataset

The synthetic data set consists of four variables, parameterised by a single variable. The function is adapted from the two-dimensional synthetic data set used in an open source tutorial (available at https://github.com/oduerr/dl_tutorial/blob/master/tensorflow/vae/vae_demo-2D.ipynb). The function generates samples of the single parameter z from the uniform distribution between 0 and 1:

$$z \sim \mathcal{U}(0,1) \quad (7)$$

Further parameters are derived from z :

$$r = \sqrt{z} \quad (8)$$

$$\eta = 0.25 \times \pi \times z$$

The mean values of each of the four variables are derived from these parameters as follows:

$$x_{1,\mu} = r \times \cos \eta \quad (9)$$

$$x_{2,\mu} = 5r \times \sin \eta$$

$$x_{3,\mu} = 2r \times \tan \eta$$

$$x_{4,\mu} = 3 \times e^{-\eta}$$

The four variables are then sampled from Gaussian distributions as follows:

$$x_1 = \mathcal{N}(\mu = x_{1,\mu}, \sigma^2 = 0.07 \times z^2) \quad (10)$$

$$x_1 = \mathcal{N}(\mu = x_{2,\mu}, \sigma^2 = 0.05 \times z^2)$$

$$x_1 = \mathcal{N}(\mu = x_{3,\mu}, \sigma^2 = 0.06 \times z^2)$$

$$x_1 = \mathcal{N}(\mu = x_{4,\mu}, \sigma^2 = 0.04 \times z^2)$$

10000 datapoints were generated using the equations above.

Simulated milling circuit dataset

The milling circuit data is based on the simulation described in (le Roux et al., 2013), with disturbances added in (Wakefield et al., 2018), and is shown in Fig. 2. The milling circuit has three control loops, and the circuit responds to stochastic variations (representing disturbances) in the simulated ore size distribution and grindability, as well as incorporating measurement error.

The original data from the simulation consists of samples every 30 seconds, for 104 days. The data has been subsampled to a sample every 15 minutes, in order to introduce some independence between samples (a key assumption in most data imputation approaches). This resulted in a set of 9985 datapoints.

3.5 Missing values

Missing data was introduced to the data in two manners, resulting in a lightly corrupted dataset, and a heavily corrupted dataset. The uncorrupted datasets were used to calculate the accuracy of the imputation methods.

The lightly corrupted dataset was created by randomly selecting 20% of the rows of a dataset, and replacing a randomly selected variable in each of these rows with NaN (not a number).

Further missing values were introduced by randomly selecting 50% of the rows of the lightly corrupted datasets, and replacing a randomly selected variable in each of these rows with NaN. This procedure was repeated three times, resulting in highly corrupted datasets. As shown in Table 1, less than 10% of each dataset was complete (and therefore available for training), about 80% of the data had one or two missing values, and about 10% of the data had three or four missing values.

3.6 Software and hardware

All code was implemented in Python 3.6, on a consumer-grade laptop (Dell Latitude E5570). The variational autoencoder code was adapted from an open source implementation (available at <https://github.com/altosaar/variational-autoencoder>).

4 RESULTS

4.1 Hyperparameters

For PCA imputation, the number of retained principal components was defined in terms of the retained variance,

which was specified at 99%. The maximum imputation iterations was set at 1000 to ensure convergence of PCA imputation.

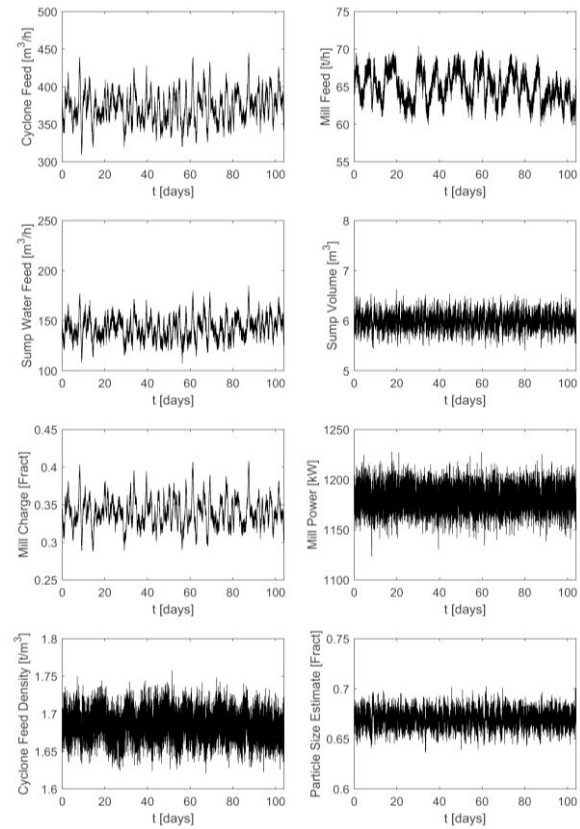


Fig. 2. Simulated milling circuit dataset

Table 1. Heavily corrupted synthetic and simulated milling circuit datasets: complete records and records with missing values

Missing values per row	Rows in synthetic dataset	Rows in milling circuit dataset
0	980	974
1	4299	3830
2	3929	3810
3	771	1273
4	20	97

For VAE imputation, the encoder and decoder were fully connected networks with two hidden layers, each with 20 neurons per layer, using rectified linear unit activation functions. The latent space was 10-dimensional. Training was run for 10 000 iterations of batches of 250 samples, with the RMSProp optimiser and a learning rate of 0.001. The maximum imputation iterations was set at 25, as initial experiments indicated convergence within 25 iterations.

4.2 VAE training

The variational autoencoder was trained on the complete records in each dataset, in order to determine the network weights which maximise the evidence lower bound (ELBO) in

equation (6). The convergence of the ELBO for the heavily corrupted synthetic dataset is shown in Fig. 3; similar (but less noisy) convergence was achieved for the lightly corrupted synthetic dataset. For the milling dataset, the ELBO converged to lower values than the synthetic dataset, suggesting that the VAE was able to extract less structure from the simulated data, likely due to the effects of simulated measurement noise and process dynamics.

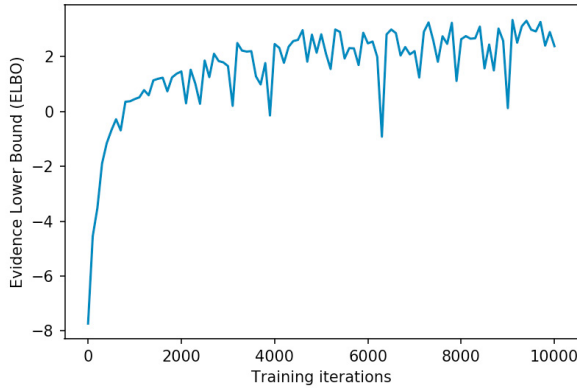


Fig. 3. Evidence lower bound (ELBO) for heavily corrupted synthetic dataset. ELBO converges after approximately 10 000 iterations.

Sample output of the input data and generated data for the lightly corrupted synthetic dataset, after 10 000 iterations, is shown in Fig. 4. The VAE successfully reproduces the non-Gaussian distributions and nonlinear relationships between the four variables; similar results are achieved for both datasets, for both levels of corruption.

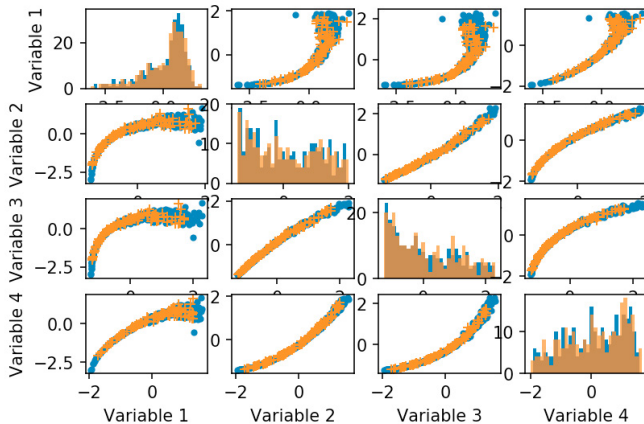


Fig. 4. Comparison of lightly corrupted synthetic data (blue dots) with data generated by the VAE (orange pluses), after 10 000 training iterations. Variable histograms are shown on the diagonal, and scatter plots of variable pairs on the off-diagonals.

4.3 Imputation results

The root-mean-square error (RMSE) was calculated for the imputed values, with the uncorrupted datasets being used as the ground truth. Each dataset was standardised with respect to the uncorrupted dataset, in order to ensure fair comparisons of the RMSE values for different levels of corruption.

The results for the four datasets and three imputation methods are shown in Table 2.

For the synthetic dataset, mean replacement appears to be a poor strategy; a significant reduction in RMSE is attained for PCA imputation, which takes into account more information about the data than simple mean replacement. However, VAE imputation achieves an order of magnitude reduction in RMSE for the lightly corrupted data, as it accounts for the nonlinearities in the data set. Although RMSE for VAE imputation doubles for the heavily corrupted dataset, it is still better than the other methods.

For the simulated milling circuit dataset, mean replacement appears to be a successful strategy; this is because the dataset represents relatively stable operation of a chemical plant under feedback control, with some fluctuation about the mean.

PCA imputation performs significantly worse than mean replacement, which is unexpected, as the variables should have approximately linear relationships (due to controlled operation around a set point). The impact of noise in the data, and the variance retained, on PCA imputation may require more detailed investigation.

VAE imputation had the lowest RMSE for the lightly corrupted dataset, but was not much better than mean replacement for the heavily corrupted dataset. This result suggests that there is some structure in the data which can be learnt by the VAE, if sufficient data is available (as in the lightly corrupted set). Where limited data is available, as in the heavily corrupted dataset, simpler, lower-variance models such as mean imputation may be more appropriate than lower-bias (but higher-variance) approaches, such as the VAE, or even PCA.

Table 2. Root-mean-sum of errors for imputation methods on the four datasets.

Dataset	Corruption type	Imputation method		
		Mean	PCA	VAE
Synthetic	Light	0.851	0.253	0.0772
	Heavy	0.844	0.279	0.160
Milling circuit	Light	0.825	1.52	0.750
	Heavy	0.809	1.21	0.796

5 CONCLUSIONS

The results above demonstrated the imputation of missing data by three approaches: single imputation by mean replacement, and multiple imputation by principal component analysis imputation and by variational autoencoder imputation. The methods were compared using a synthetic nonlinear dataset, and a simulated milling circuit dataset which more closely resembles real industrial data.

The training of the VAE showed that the method can successfully learn distributions in latent space representations of complex, multimodal, nonlinear datasets in an unsupervised manner, even in the presence of heavy corruption (~10% complete records).

VAE imputation outperformed the other two methods, on both lightly and heavily corrupted datasets, as it can account for complex and nonlinear variable relationships and multimodal variable distributions. Further detailed investigation into the effects of noise and retained variance in PCA imputation should be conducted, to confirm the results presented here.

Inferential sensing is a promising area for future VAE applications, particularly in minerals processing, in which critical parameters (such as composition and particle size) are measured offline. Some process knowledge may be required in order to correctly structure the training data (for example, grab samples may be aggregated over shifts), but the extension of the demonstrated missing data imputation to inferential sensing should be relatively simple, and may be able to take advantage of more recent advances in the use of variational autoencoders in semi-supervised learning problems (Kingma et al., 2014)

6 ACKNOWLEDGEMENTS

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